

Marcio Merlone

Senior Software Engineer | Full-Stack, Web Applications & Systems Architecture

Araucária / Curitiba, PR, Brazil | Email: mmerlone@gmail.com | LinkedIn: [linkedin.com/in/mmerlone](https://www.linkedin.com/in/mmerlone) | Portfolio: mmerlone.dev.br

PROFESSIONAL SUMMARY

Results-driven **Senior Software Engineer** with over 20 years of technical expertise, specializing in building scalable, resilient web applications with a systems engineer's perspective. Proven track record at **ArcTouch** delivering high-visibility, WCAG-accessible React/TypeScript solutions for global brands (HP, Cirrus Aircraft, Quizlet, Starbucks) and architecting complex headless CMS migrations. Uniquely bridges modern full-stack engineering (React 19, Next.js 15, Node.js, GraphQL, Supabase, Vercel, Tailwind, MUI v7) with deep infrastructure mastery (Linux, VMware virtualization, Proxmox, cloud migrations, network security). Passionate about clean architecture, type safety, observability, and long-term system reliability.

TECHNICAL SKILLS & CORE COMPETENCIES

Frontend Architecture:	React 19, TypeScript, Next.js (15+ App Router), Material UI (MUI v7), TailwindCSS, Redux, Responsive Design, WCAG 2.1 Accessibility, HTML5/CSS3
Backend & Cloud Services:	Node.js, GraphQL, Supabase (Postgres, RLS), Vercel, REST APIs, PHP, Perl, Microsoft 365 / Exchange Online, Hybrid Cloud, Apache, Postfix, Dovecot, OpenLDAP
DevOps & Infrastructure:	Docker, Proxmox, Git, GitLab (Self-Hosted), CI/CD Pipelines, pfSense Firewall, VLAN Segmentation, IPsec/OpenVPN, VMware ESXi/vCenter, Zabbix, Graylog
Testing & Quality:	Jest, Cypress, React Testing Library, Zod, React Hook Form, React Query, Sentry, Pino Logging, Code Review, Performance Profiling
Practices & Governance:	Agile / Scrum Rituals, Product Ownership, Identity & Access Management (IAM, RBAC), IPAM, Systems Migration, Infrastructure Hardening, ISO 9001 Compliance

PROFESSIONAL EXPERIENCE

Senior Software Engineer

March 2022 – Present (4+ yrs) | Remote

ArcTouch – High-Impact Mobile & Web Solutions for Global Brands

- **Frontend & React Engineering:** Engineered scalable, performant web applications for global enterprise clients (HP, Cirrus Aircraft, Quizlet, Starbucks) utilizing React, TypeScript, Next.js, and component-driven architecture.
- **WCAG Accessibility Leadership:** Championed WCAG 2.1 accessibility standards across client projects, implementing semantic HTML, keyboard navigation, and ARIA patterns to ensure inclusive user experiences.
- **Headless CMS Migration & Authoring:** Architected and executed custom data extraction/transformation scripts to migrate Cirrus Aircraft from legacy WordPress to Contentstack CMS and Discourse platform. Authored the published engineering case study on ArcTouch's tech blog.
- **SaaS Platform Customization:** Extended leading SaaS platforms by developing custom Shopify storefront themes (Liquid) and Discourse community plugins (Ruby on Rails, Ember/Handlebars).
- **Testing & Code Quality:** Standardized testing procedures by writing comprehensive unit and integration test suites (Jest, Cypress, React Testing Library) and leading rigorous peer code reviews.

IT Staff / Network & Systems Manager

March 2007 – March 2022 (15 yrs) | Curitiba, Brazil

A1 Tecnologia Industrial – Engineering & Industrial Solutions

- **Infrastructure Scaling & Lifecycle Management:** Scaled enterprise IT operations from ~60 to 1,000+ endpoints supporting >400% organizational expansion across headquarters, remote branch offices, and distributed teams.
- **Virtualization & Server Consolidation:** Consolidated 15 legacy physical servers into a high-availability VMware ESXi/vCenter virtualized environment with 60+ VMs and redundant SAS storage, significantly lowering CapEx.
- **Cloud & Identity Migration:** Directed seamless migration of 300+ users to Microsoft Cloud (Exchange Online) and upgraded network identity from legacy LDAP to Active Directory with zero operational downtime.
- **Network Security & Segmentation:** Hardened perimeter security by deploying virtualized pfSense firewalls, implementing strict VLAN segmentation, IPsec VPN tunnels, and centralized Graylog/Zabbix observability.
- **DevOps Modernization & Governance:** Spearheaded DevOps modernization by deploying an in-house self-hosted GitLab server to replace deprecated CVS repositories. Led IT governance and ISO 9001 compliance auditing.

Full-Stack Software Developer & Systems Analyst Roles

2001 – 2007 | Curitiba & São Paulo, Brazil

Áton Tecnologia, AmbienteBrasil, Locaweb Ltda, Dialdata Ltda

- **Full-Stack Development (Áton & AmbienteBrasil):** Built custom PHP MVC web frameworks, Smarty templating systems, AJAX interfaces, and managed environmental web portal servers (Apache, DNS, Postfix, Wireless links).
- **High-Volume SaaS Platforms (Locaweb & Dialdata):** Maintained user administration and webmail portals for major web hosts using Perl, PHP, and MySQL. Engineered resilient Postfix/OpenLDAP mail clusters on FreeBSD, Solaris, and Linux.

- **Furukawa Industrial (Mechanical Design Drafter / Production):** Executed 2D/3D CAD design (AutoCAD, Solid Edge), production line installations, supplier negotiations, and ISO quality auditing in manufacturing plants (Bosch, Volvo, Siemens). Developed foundational physical systems discipline.
- **Early Systems & Consulting:** Deployed early Linux network servers (Squid proxy, dial-up ISDN, file sharing) and delivered Lotus/IBM enterprise software technical support and training.

FEATURED ENGINEERING PROJECTS & OPEN SOURCE

YwyBase — Solid Ground to Scale (Next.js 15+ & React 19 Starter)
Designed and open-sourced a production-grade full-stack starter featuring Supabase Auth + Postgres, Vercel, MUI 7 + Pigment CSS, Zod, React Query, Sentry observability, Pino logging, and security middleware. Demo: [ywybase.vercel.app](#)

@mmerlone/mui7-phone-number (Published npm Package)
Created and maintain an open-source phone number input component for Material UI v7+ featuring auto-formatting, country selection, and strict TypeScript definitions. Live demo and npm package maintainer.

@mmerlone/react-tz-globepicker (Published npm Package)
Developed an interactive 3D globe component for React 19 applications requiring timezone selection and compact world-time visualization with zero external runtime overhead.

Headless CMS Migration Case Study (Published Engineering Article)
Authored an in-depth engineering article published on ArcTouch's blog detailing automated data extraction and transformation pipelines migrating legacy WordPress data into Contentstack and Discourse.

EDUCATION & PROFESSIONAL DEVELOPMENT

Associate Diploma in Mechanical Engineering Technology	CEFET-PR (Centro Federal de Educação Tecnológica do Paraná)	1986 – 1989
Undergraduate Coursework in Informatics (2 Years Completed)	Universidade Federal do Paraná (UFPR)	1993 – 1994
Undergraduate Coursework in Computer Science (2 Years Completed)	Pontifícia Universidade Católica do Paraná (PUCPR)	1991 – 1992
Certifications & Continuous Education: Scrum: The Basics (Agile Methodologies) ISO 9001:2008 Internal Auditor Microsoft Certified Professional (MCP #2432097) Advanced Specializations: React Server-Side Rendering (SSR), React Design Patterns & Accessibility (WCAG), Building AI Applications with React & OpenAI API.		